
The Pro-Action Operator: A Scientifically-Informed, Multi-Timescale Self-Regulation Model for LLM-Agents

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Abstract

Contemporary LLM-agent frameworks — ReAct, LangGraph, OpenClaw — have prioritized *capability* over *motivation*, deciding *when* an agent acts through exogenous flow control or task-backlog heartbeats. Homeostatic reinforcement learning (Keramati–Gutkin 2014) and active-inference agents (Friston 2017; Tschantz et al. 2020) internalize regulation, but often through a compact objective rather than an explicit set of delayed regulatory subsystems. However, biological regulation is not single-timescale: converging work in stress physiology, self-regulation, affect, and cognitive control treats adaptation as the interaction of subsystems — attention, perception, hormonal (SAM adrenergic, seconds; HPA cortisol, minutes), emotional, neuropsychological, cognitive — exchanging signals across distinct delays, consistent with the mechanism behind Yerkes–Dodson inversion and delayed reappraisal (Porcelli–Delgado 2017). We propose the **Pro-Action operator** Γ , a recursively composable, multi-timescale thermostat system. Each subsystem k updates as

$$x_{k,t+1} = x_{k,t} - \kappa_k(x_{k,t} - x_{k,t}^*) + \lambda_k - \alpha_k \rho_k(a_t, e_{t+1}) + [W \phi(\mathbf{x}_{t-\tau_k})]_k,$$

with $\phi = \tanh$ elementwise, subsystem-specific delay τ_k encoding a dimensionless SAM–HPA-like ratio ($\tau_{\mathcal{H}}^{\text{HPA}} \gg \tau_{\mathcal{A}}$), and action selected by an arousal-gated mixer. Γ is proposed as a finite, inspectable parameterization of regulation: the τ_k are explicit hyperparameters rather than hidden in orchestration depth. We specify three falsifiable properties a priori: (P1) elastic return to set-point; (P2) Yerkes–Dodson inverted-U; (P3) delayed reappraisal. As a Concept & Feasibility contribution, the paper reports the model, a lightweight falsifiability protocol, computational sanity checks, and a preliminary LLM-agent benchmark snapshot. The current evidence is not a final validation; it shows that the operator can be coupled to frontier LLM agents while preserving non-degenerate, opponent-conditional behavior under partial coverage — a feasibility claim, not a causal claim about the operator’s contribution over baselines.

Keywords. Homeostatic reinforcement learning; LLM agents; active inference; multi-timescale self-regulation; endogenous activation; cybernetics; self-regulation model.

1 Introduction

The deployment of LLM-based autonomous agents into settings of institutional consequence — health triage, financial advice, public deliberation — is moving faster than the science of how such agents should *regulate themselves*. Whether an agent speaks, waits, defers, or escalates tends to be decided outside the agent, by an orchestrator whose control rules may be less expressive than the

33 situations they coordinate: a small set of graph transitions, queues, or heartbeats is used to govern
34 high-dimensional and shifting social or task states. In Ashby’s terms, the controller may lack the
35 requisite variety needed to match the variety of the environment [1].

36 A frequently reported symptom, consistent with this variety mismatch, is the loss of activation
37 coherence in adversarial multi-agent deliberation: floor monopolization, context collapse, and
38 interventions that appear decoupled from any internal state the agent could be said to carry (evidence
39 observed by the authors in a companion work, currently under review). The pattern suggests that
40 *when* an agent acts may depend on structure the dominant paradigm does not explicitly represent.

41 Contemporary LLM-agent frameworks — ReAct, LangGraph, OpenClaw — have prioritized *ca-*
42 *pability* (what an agent can do) over *motivation* (why and when it should act at all), addressing
43 activation via cognitive while-loops and exogenous flow control. Homeostatic reinforcement learning
44 (Keramati–Gutkin 2014) and active-inference agents (Friston 2017; Tschantz et al. 2020) internalize
45 regulation, but do not provide a compact LLM-agent operator in which attention, interoception,
46 emotion, memory, cognition, and stress-like physiology are all explicit delayed regulators. Intelligent
47 animals, including humans, regulate behavior through multiple interacting systems — attention,
48 arousal, affect, memory, deliberation — operating on distinct timescales whose relative priority
49 shifts with context; dominant LLM-agent paradigms treat activation as a single-objective signal or
50 exogenous flow control.

51 The problem is older than LLM-agents. Mid-20th century behaviorism defined it precisely by
52 bracketing it: the stimulus–response mapping could be studied scientifically because the intervening
53 machinery — the *black box* — was deliberately left unspecified. Decades of subsequent research
54 opened that box from different angles: behavioral biology established that animal regulation is
55 multi-drive and hierarchical; stress physiology revealed fast (adrenergic, seconds) and slow (cortisol,
56 minutes) adaptation axes with distinct roles; affective neuroscience showed that emotion is not
57 an output label but a construction that shapes perception, memory, and action; evolutionary game
58 theory demonstrated that social behavior is governed by reciprocity and temporal discounting, not
59 instantaneous utility; cybernetics formalized why a regulator must match the variety of what it
60 controls. Each tradition contributed a piece. What has not been done for LLM-agents is assembling
61 them into a single compact, falsifiable operator. Section 2 reviews each contribution in detail.

62 What remains comparatively unexplored is a single computable, falsifiable operator that integrates
63 these disciplinary insights for LLM-agents. The central biological clue is that regulation is not a
64 single controller but a *system of subsystems* — attention, perception, hormonal (SAM adrenergic,
65 seconds; HPA cortisol, minutes), emotional, neuropsychological, cognitive — exchanging signals
66 across distinct delays. The SAM–HPA timescale split is not decorative: it is consistent with the
67 mechanism behind Yerkes–Dodson inversion, post-stress reappraisal windows, and delayed-valuation
68 shifts [2]. The scientific question is therefore whether activation itself can be treated as a regulated
69 internal process — a coupled set of thermostat-like variables whose delayed interactions determine
70 when the agent should speak, wait, defer, or escalate — and whether the smallest such structure
71 remains distinguishable from a single-timescale regulator.

72 We propose the **Pro-Action operator** Γ as a model, not an experimental technique. Γ treats activation
73 as a recursively composable system of six coupled thermostats with explicit update delays. The
74 contribution type is Concept & Feasibility: the idea is intentionally broader than what can be validated
75 in a single benchmark, but it must still be made operational enough to fail. The contributions of
76 this paper are primarily *modeling*: (i) a formal operator integrating multi-timescale regulation; (ii)
77 the framing of Γ as a Self-Regulation Model (SRM) class complementary to capability-oriented
78 LAMs; (iii) an a priori specified falsifiability protocol; (iv) a reference implementation checked by
79 computational sanity tests; and (v) a preliminary LLM-agent benchmark snapshot used as feasibility
80 evidence rather than final empirical adjudication.

81 2 Related work

82 **LLM-agent orchestration.** Recent LLM-agent frameworks such as ReAct [3], LangGraph, and
83 OpenClaw made language models operational by embedding them in action loops, tool graphs,
84 and multi-agent workflows. This was a necessary engineering step: without explicit structure, an
85 LLM has no durable procedure for acting in an environment. More recently, agent architectures
86 have incorporated persistent memory streams, reflection, and simulated affective states to enrich the

87 content of decisions [4–7]. The limitation common to both generations is that activation is usually
 88 externalized: the graph or supervisor decides when the model acts; the model supplies content for
 89 that externally selected moment. Γ is complementary: those works provide richer *content*; Γ provides
 90 a principled internal account of *timing*.

91 **Internal regulation across timescales.** A different tradition begins from the opposite assumption:
 92 action should be driven by internal regulatory state. Homeostatic RL makes internal deficit part of
 93 the reward [8]; active inference gives a broader variational formulation [9, 10]. These approaches
 94 are important — they move regulation inside the agent — but do not yield a compact operator
 95 with separately inspectable subsystem delays. Once regulation is internalized, the next question
 96 is whether one regulatory signal is enough. Stress physiology and allostasis suggest it is not: fast
 97 sympathetic responses (SAM, seconds) and slower endocrine/cognitive recovery (HPA, minutes) play
 98 distinct adaptive roles [11–13]. The remaining question is which internal variables deserve to be
 99 modeled. Interoceptive and somatic-marker accounts connect bodily state to action readiness [14, 15];
 100 affect-regulation accounts treat emotion as part of how situations are appraised [16, 17]; evolutionary
 101 game theory adds reciprocity and temporal discounting as social constraints [18, 19]. Together, these
 102 literatures motivate attention, interoception, emotion, memory, cognition, and hormonal dynamics as
 103 functional regulators rather than decorative modules.

104 **Systems, cybernetics, and the gap.** If these variables are treated as separate regulators, the final
 105 problem is structural: how should they be composed? Cybernetics supplies the relevant language:
 106 Ashby’s law of requisite variety requires a regulator rich enough to match the environment [1]; Beer’s
 107 Viable System Model adds recursive organization [20]; systemism frames the relevant object as
 108 composition rather than isolated component [21, 22]. The gap might be narrow yet important. To
 109 our knowledge, the threads above — exogenous orchestration, homeostatic internalization, multi-
 110 timescale physiology, affect regulation, and cybernetic composition — have not been combined into a
 111 small, falsifiable LLM-agent operator in which the six regulatory domains are explicit state variables
 112 with explicit delays. That is the gap Γ is designed to occupy.

113 3 The Pro-Action operator

114 3.1 Axioms of homeostatic autonomy (A1–A9)

115 Before defining Γ , we make explicit the modeling commitments that the operator assumes. The
 116 axioms are not empirical results or theorems; they state what counts, in this model, as an autonomous
 117 self-regulating agent, what internal variables are allowed to regulate action, and how those variables
 118 may be composed. This is meant to avoid leaving the ontology of the agent implicit in implementation
 119 choices. The axiom set was subjected to a formal coherence check using SMT-based tools (Z3) to
 120 verify that no pair of axioms is mutually contradictory under the intended semantics; all checks
 121 passed.

122 We extend the Axioms of Homeostatic Autonomy from the companion work (A1–A3) with six
 123 additional axioms that make the multi-timescale structure explicit:

124 **A1–A3 (Kernel).** Autonomy, Impulse, and Quality Gate: the agent maintains an internal deficit δ_i
 125 that drives action probability $p_i(t) = \sigma(D(\delta_i(t)))$, reduced by action quality g .

126 **A4 (Saliency as gate).** Not every observation enters the context. An attention operator $\mathcal{A}$ selects
 127 $\tilde{o}_t = \mathcal{A}(o_t, \mathcal{I}_t, \mathcal{D}_t, \mathcal{M}_t; x_{\mathcal{A},t})$.

128 **A5 (Interoception as parallel input).** A body signal $\mathcal{I}_t$ feeds into $\mathcal{E}$ and $\mathcal{A}$ while bypassing $\mathcal{P}$ as a
 129 bottleneck.

130 **A6 (Memory as ubiquitous prior).** A state $\mathcal{M}_t$ biases all subsystems and updates post-action.

131 **A7 (Drives as set-point generators).** Set-points are not constants: $\mathbf{x}_t^* = \mathcal{D}_t$.

132 **A8 (Fast/slow parallelism with arbitrator).** Two routes produce action candidates c^{fast} , c^{slow} ; an
 133 arbitrator assigns precision weights π_{fast} , π_{slow} .

134 3.2 The operator

135 Γ is defined as the composition:

$$\Gamma := \mathcal{O} \left[\text{ARB}_\pi \left(\underbrace{(\mathcal{N}_{\text{fast}} \circ \mathcal{H}_{\text{SAM}})}_{\text{fast route}} \parallel \underbrace{(\mathcal{C} \circ \mathcal{E} \circ \mathcal{P})}_{\text{slow route}} \right) \circ \mathcal{A} \right] \Big|_{\mathcal{I}, \mathcal{M}, \mathcal{D}} \quad (1)$$

136 Each subsystem k updates as:

$$x_{k,t+1} = x_{k,t} - \kappa_k(x_{k,t} - x_{k,t}^*) + \lambda_k - \alpha_k \rho_k(a_t, e_{t+1}) + [W \phi(\mathbf{x}_{t-\tau_k})]_k \quad (2)$$

137 with $\phi = \tanh$ elementwise. Action is selected by:

$$a_{t+1}^* = \arg \min_a \pi_f \mathcal{F}_{\text{short}} + (1 - \pi_f) \mathcal{F}_{\text{long}} + \gamma \|Q^{1/2}(\mathbf{x}_{t+1}(a) - \mathbf{x}_{t+1}^*)\|^2, \quad \pi_f = \sigma(c \cdot x_{\mathcal{H}} + b) \quad (3)$$

138 where $\mathbf{x}_{t+1}(a)$ denotes the predicted next state under candidate action a , depending on a through the
 139 action-quality term $\rho_k(a, e_{t+1})$ in Eq. (2) — it is this dependence that makes the homeostatic deficit
 140 penalty action-relevant. $\mathcal{F}_{\text{short}}, \mathcal{F}_{\text{long}}$ are generic cost functionals inspired by expected-free-energy
 141 control over horizons $h_s \ll h_l$, and $Q \succeq 0$ is a diagonal precision matrix (distinct from the coupling
 142 matrix W in Eq. (2)). In the IPD instantiation below we do not claim a strict active-inference
 143 implementation with an explicit generative model, and the mixer is not literally executed as an argmin
 144 over LLM completions: the controller modulates the prompt seen by the LLM (Section 4.3) and the
 145 LLM emits the action; Eq. (3) is the compact controller-level expression of fast/slow arbitration that
 146 the prompt encodes.

147 3.3 Non-absorption of delay parameters

148 **Proposition 1** (Non-absorption of delays). *Let Γ_τ denote Eq. (2) with delays $\tau = (\tau_1, \dots, \tau_6) \in \mathbb{Z}_{\geq 0}^6$
 149 and coupling matrix $W \neq \mathbf{0}$. For any two delay vectors $\tau \neq \tau'$ with $\tau_k \neq \tau'_k$ for at least one k such
 150 that $[W]_{k,\cdot} \neq \mathbf{0}$, there is no reparametrization $(\kappa'_k, \lambda'_k, \alpha'_k)$ of the non-delayed terms that makes the
 151 impulse-response of $\Gamma_{\tau'}$ identical to that of Γ_τ . Equivalently, τ_k is not absorbable into $(\kappa_k, \lambda_k, \alpha_k)$.*

152 The argument (Appendix A) is an order-mismatch argument in the discrete-time z -domain: the
 153 delayed subsystem has transfer function $1/(z^{\tau_k+1} - (1 - \kappa_k)z^{\tau_k})$, of denominator degree $\tau_k + 1$,
 154 with an impulse response identically zero for the first τ_k samples by causality. A first-order delay-
 155 free reparametrization cannot enforce this delayed zero-output period. Two systems with distinct
 156 delays therefore produce distinct cross-correlation peaks (the first non-zero lag is $\tau_k + 1$), making
 157 τ_k identifiable from observed trajectories. This is not a structural identifiability theorem for the full
 158 nonlinear system (which remains future work), but it establishes that the delays are non-redundant
 159 parameters and that Γ occupies a distinct model class from any delay-free parametrization of the
 160 same dimension.

161 3.4 Reductions to simpler models

162 **Reduction to Driveplexity.** Driveplexity (companion work, currently under review) is a single-drive
 163 homeostatic activation model in which each LLM-agent maintains a scalar epistemic deficit δ_i that
 164 decays at a fixed rate and is reduced by the quality of each action executed. Γ subsumes it: when
 165 $x_{\mathcal{A}} = \dots = x_{\mathcal{N}} \equiv 0$, $\mathcal{A} = \text{id}$, $\mathcal{I} = \mathcal{M} = \mathcal{D} = \emptyset$, $\pi_{\text{fast}} \equiv 0$, $W = \mathbf{0}$, $\kappa_{\mathcal{C}} = 0$, and $\mathbf{x} \rightarrow x_{\mathcal{C}} \equiv \delta_i$,
 166 Equation (2) reduces to $\delta_{t+1} = \delta_t + \lambda - \alpha \cdot g(e^{(t)}, e^{(t+1)})$, which is exactly A1–A3.

167 **Reduction to homeostatic RL.** When $\kappa = \mathbf{0}$ and all $\tau_k = 0$, Γ becomes a reactive drive-based policy
 168 consistent with the simplified form of Keramati–Gutkin (2014).

169 4 Falsifiability protocol

170 To avoid the vacuity of untestable models, three properties are specified a priori before any experiment:

171 **(P1) Elastic return to set-point.** Under perturbation, each x_k returns to x_k^* with measurable half-life
 172 $t_{1/2}$.

173 **(P2) Yerkes–Dodson inverted-U.** Arousal vs. performance follows an inverted-U; ablation of SAM
 174 coupling ($\mathcal{H}_{\text{SAM}}$) collapses it to monotone. Target BFI ≥ 0.70 against Porcelli–Delgado (2017).
 175 **(P3) Delayed reappraisal.** Cross-lag correlation between $x_{\mathcal{C}}$ and $x_{\mathcal{E}}$ peaks at lag $\tau_{\mathcal{H}}^{\text{HPA}}$.

176 4.1 Behavioral Fidelity Index (BFI)

$$\text{BFI} = \exp(-W_1(\mu_{\text{agent}}, \mu_{\text{human}})/\lambda) \quad (4)$$

177 over normalized marginals of cooperation rate, action volatility, and log-response-time proxy. The
 178 response-time proxy is an engineering observable, not a claim of equivalence to human decision
 179 time. The model is supported if BFI exceeds thresholds on ≥ 2 of 3 properties and the corresponding
 180 ablation degrades the property.

181 4.2 Experimental design

182 We test Γ within the Iterated Prisoner’s Dilemma (IPD). The reason for this choice is that IPD has
 183 extensive empirical evidence of human behavior in repeated social interaction [23, 18, 2], providing a
 184 non-trivial baseline for comparing agent behavior against human patterns. The setting we establish
 185 for the game is 10% noise and 50 rounds, with a forced-defection perturbation at round 10 to test
 186 the elastic-return property (P1) by creating a measurable deviation from set-point. Sotopia-derived
 187 deliberation [24] is deferred to future work because it is better suited for ecological validity than for
 188 the first controlled falsification of the operator.

189 Baselines serve two purposes: ReAct and LangGraph with memory represent the dominant exogenous-
 190 activation paradigm; Driveplexity/homeostatic-RL reduction tests whether Γ adds value beyond a
 191 single-drive controller; a small discrete active-inference comparator implemented in pymdp checks
 192 whether the fast/slow arbitration in Γ captures a distinct signature from standard active inference.

193 Three instruction-tuned LLM providers are used as policy-execution layers: OpenAI (gpt-5-nano),
 194 Anthropic (claude-haiku-4-5), and DeepSeek (deepseek-v4-flash). These models were se-
 195 lected as the comparable lite/flash/nano variants of each provider’s frontier line — well-benchmarked
 196 and roughly matched in capability tier. Using three independent providers reduces dependence of the
 197 controller-level findings on any single provider’s instruction-tuning regime. Each provider/model is
 198 held fixed across conditions, so that within-provider contrasts isolate the controller rather than model
 199 identity. Provider-level summaries are reported as diagnostics, not as claims about provider quality.

200 **Inference parameters.** OpenAI gpt-5-nano uses default temperature with a per-round seed
 201 for reproducibility; Anthropic claude-haiku-4-5 uses temperature 0 and max_tokens 512 (no
 202 seed parameter exposed); DeepSeek deepseek-v4-flash uses temperature 0 and max_tokens
 203 512. Calls return JSON objects; OpenAI and DeepSeek use response_format: json_object.
 204 Concurrency is capped at 8 in-flight cells per provider, with up to 4 exponential retries on transient
 205 errors.

206 All frozen constants are assigned a provenance category before evaluation (Table 1). Values marked
 207 as smoke-test calibration were chosen before the LLM benchmark to satisfy non-divergence, set-point
 208 return, and visible perturbation response in the numerical simulator; final reporting will include
 209 sensitivity checks for critical constants.

Table 1: Frozen parameter provenance for the IPD instantiation.

Parameter	Value	Provenance
λ	[.08, .07, .10, .08, .06, .10]	smoke-test calibration
α	[.20, .15, .25, .20, .10, .15]	smoke-test calibration
κ	[.10, .10, .15, .12, .08, .10]	smoke-test calibration
$\mathbf{x}^*$	[.3, .2, .3, .1, .2, .4]	principled prior / stable interior set-point
$\tau_{\text{SAM}}, \tau_{\text{HPA}}$	1, 5 rounds	dimensionless fast/slow ratio
W	sparse signed coupling matrix	functional constraint and ablation interpretability
Q	I_6 (identity)	smoke-test calibration; precision matrix in Eq. (3)
Rounds	50	comparability requirement for final cells

Seven ablation conditions (Table 2) isolate the contribution of the main regulatory components. Full- Γ uses all six thermostats; No- $\mathcal{H}$ removes the fast SAM route; No- $\mathcal{E}$ neutralizes emotional valence; No- $\mathcal{N}$ removes the slow cognitive route. Random- Γ permutes the entries of W while preserving all other parameters — same six thermostats, same τ_k , same hyperparameters, but biologically-unmotivated wiring — to isolate the contribution of the coupling structure rather than parameter count. Collapse-NC merges the neuropsychological ($\mathcal{N}$) and cognitive ($\mathcal{C}$) subsystems into a single regulator by averaging their parameters and symmetrizing their W rows and columns, testing whether the 6-subsystem partition is necessary or whether 5 would suffice.

Table 2: Ablations. Full- Γ uses all six thermostats; ablations remove, permute, or collapse selected regulatory components.

Condition	Change from Full- Γ	Purpose
Full- Γ	—	Reference
No- $\mathcal{H}$	Hormonal channel ($\mathcal{H}_{\text{SAM}}$, $\mathcal{H}_{\text{HPA}}$) zeroed	Fast-timescale contribution
No- $\mathcal{E}$	Emotional valence zeroed	Affect as stabilizer vs. noise
No- $\mathcal{N}$	Neuropsychological channel zeroed	Slow deliberation contribution
Random- Γ	W entries permuted; τ_k , κ , λ , α unchanged	Biologically-motivated wiring vs. parameter count
Collapse-NC	$\mathcal{N}$ and $\mathcal{C}$ merged into one regulator	6-subsystem vs. 5-subsystem partition

4.3 Prompt protocol and label-priming control

The controller modulates the LLM through a frozen, version-controlled prompt template: each round, the six thermostat values $\mathbf{x}_t$ and the aggregated proposal p_C , h_{SAM} are inserted with semantic anchors pairing each numeric value with its set-point and a qualitative low/high interpretation (e.g., hormonal = 0.58 (set-point 0.30; low = under-aroused / calm, high = elevated stress / arousal)). The LLM may follow or override $p(C)$; the controller modulates rather than overrides. A known failure mode is that the LLM responds to *labels* rather than to the underlying dynamics. To address this, the protocol pre-specifies a *scrambled-labels* control arm in which the six semantic labels are permuted to neutral identifiers (`metric_A`, `metric_B`, ...) while numeric values remain bound to their true subsystem; behavioral identity between Full- Γ and scrambled-labels would confirm the controller dynamics carry the signal rather than lexical priming. This condition is reserved for the post-completion ancillary analysis. Full implementation details are available in the open repository cited under Reproducibility in §6.

5 Preliminary verification

Before empirical evaluation, we subjected Equation (2) and its reference implementation to computational sanity checks using symbolic algebra (SymPy), SMT-based counterexample search (Z3), arbitrary-precision arithmetic (mpmath), and numerical dynamical tests (NumPy). These checks are not formal verification of the scientific model. They test internal consistency: whether the implementation preserves the intended update structure, avoids immediate numerical divergence, and contains Driveplexity as a reduction.

Smoke-test results (all pass): (1) Set-point convergence: $\|\mathbf{x}_t - \mathbf{x}^*\| < 0.15$ after 10 rounds. (2) Elastic return half-life $t_{1/2} = 3.2$ rounds. (3) π_{fast} rises from 0.35 to 0.92 under threat. (4) 20 iterations without divergence. (5) Driveplexity is an invariant subspace of Γ ($\delta = -0.0513$).

Simulation trajectories. Figure 1 shows the six thermostat trajectories for two agents over 30 rounds. Both agents converge toward their set-points ($\mathbf{x}^* = [0.3, 0.2, 0.3, 0.1, 0.2, 0.4]$) despite perturbation at round 10 (forced defection).

Cumulative payoffs and the P1 elastic-return zoom on $x_{\mathcal{H}}$ are provided in Appendix B (Figures 3 and 4). These results verify the *internal consistency* of the dynamics, not biological or behavioral fidelity. Parameter sensitivity checks (Appendix B) confirm that P1 elastic return is robust to $\pm 20\%$ uniform scaling of κ and λ , indicating that the smoke-test calibration point is not isolated.

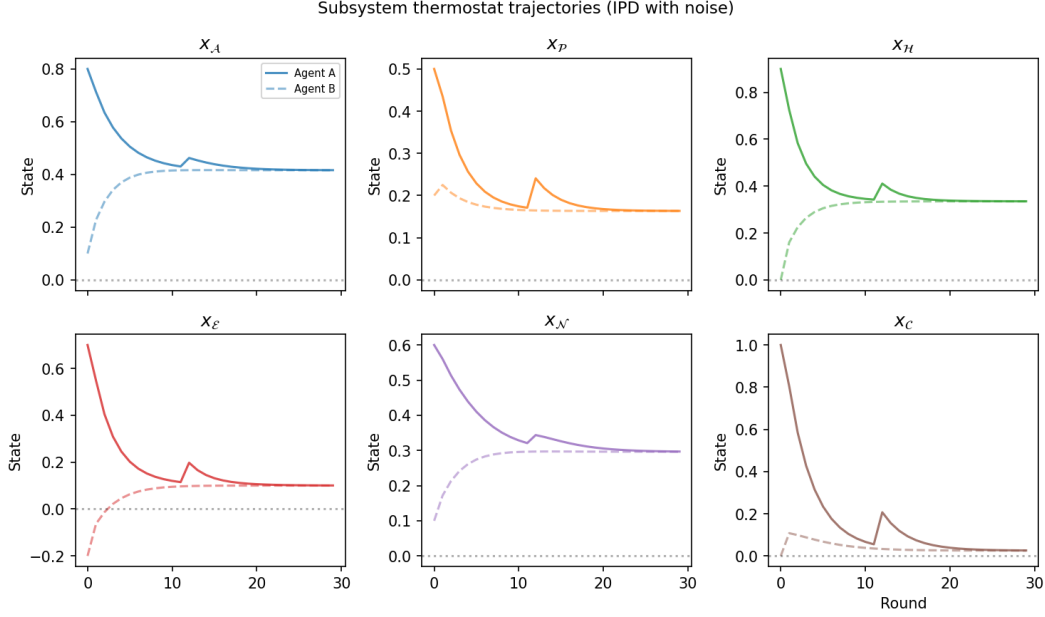


Figure 1: Subsystem thermostat trajectories for two agents in IPD with noise. The perturbation at round 10 (forced defection) triggers transient deviation, followed by elastic return toward set-points.

248 5.1 Interim LLM-agent benchmark snapshot

249 We then asked whether the operator remains meaningful when the policy-execution layer is no longer
 250 a numerical mock agent but a real LLM prompted to play Iterated Prisoner’s Dilemma under a
 251 frozen controller state. Each *cell* denotes an independent experimental unit: a tuple (Γ -condition
 252 $\times$ provider $\times$ opponent $\times$ seed) run for 50 IPD rounds. At the current snapshot, 666 cells have
 253 completed the required 50 rounds ($\approx 70\%$ of the planned grid). Five of the six conditions are at full
 254 balanced coverage: 120 Full- Γ , 120 No- $\mathcal{H}$, 120 No- $\mathcal{E}$, 120 No- $\mathcal{N}$, and 120 Random- Γ cells, each
 255 balanced across three providers and four opponent policies. Collapse-NC has 66 cells across three
 256 opponents (GTFT not yet covered). The exogenous-activation baselines (ReAct, Driveplexity) remain
 257 at zero completed cells. The results below are consequently feasibility evidence, not final empirical
 258 adjudication.

259 The first useful sign is that Full- Γ does not collapse to a trivial policy. Mean cooperation is 0.720 (sd
 260 0.165, 95% CI [0.690, 0.750]; 10k bootstrap resamples), with action volatility of 16.0 switches per 50-
 261 round cell (95% CI [15.0, 17.0]). Opponent sensitivity is preserved and notably discriminative: Full- Γ
 262 cooperates more with TFT (0.829) and GTFT (0.839), less with Grim (0.495), and intermediately
 263 with Random (0.719). The contrast with Random- Γ (now 120 balanced cells) is informative: a
 264 randomly-wired controller cooperates uniformly across all opponents — TFT 0.831, GTFT 0.825,
 265 Grim 0.843, Random 0.835 — producing nearly flat opponent sensitivity (range 0.019) compared to
 266 Full- Γ ’s sharply differentiated profile (range 0.344). The biologically-motivated coupling structure
 267 in W appears to contribute not to the *level* of cooperation but to the *discriminability* of the policy
 268 across opponent types. This is the strongest internal evidence so far that the wiring of Γ matters
 269 beyond mere parameter count. We are explicit that this does *not* establish that the controller *causes*
 270 the discrimination — that contrast requires baselines (ReAct, Driveplexity) that are not yet covered.

271 The ablations expose a differentiated picture. No- $\mathcal{H}$ is close to Full- Γ in cooperation (0.745 vs. 0.720,
 272 95% CI difference overlapping) but meaningfully lower in volatility (14.1 vs. 15.9 switches). The
 273 cooperation-rate similarity does not refute the hormonal channel’s contribution; it re-focuses the
 274 question onto dynamic responsiveness under perturbation rather than baseline levels. No- $\mathcal{E}$ is nearly
 275 indistinguishable from Full- Γ in aggregate cooperation (0.715 vs. 0.720) and volatility (15.9 vs.
 276 15.9), suggesting that under normal conditions emotional valence is neither beneficial nor disruptive
 277 — but the degenerate-regime evidence below qualifies this. No- $\mathcal{N}$, now with 120 cells and full

opponent balance, shows cooperation essentially equal to Full- Γ (0.722 vs. 0.720, CIs overlapping); the apparent cooperation gap of 0.054 in earlier 68-cell data (GTFT missing) has closed entirely with balanced coverage, validating the coverage-bias interpretation. Notably, No- $\mathcal{N}$ has the shortest recovery delay (6.42 rounds, 95% CI [5.42, 7.42], vs. Full- Γ 7.23 [6.38, 8.10]). This is consistent with the asymmetric prediction: $\mathcal{H}$ acts on the fast timescale ($\tau_{\text{SAM}} = 1$) and its removal should *slow* recovery; $\mathcal{N}$ acts as the slow deliberative channel and its removal should *accelerate* recovery by reducing slow-cycle inertia — the two ablations predict opposite-sign effects on the same observable, which is precisely what pre-committed criterion (i) below tests.

To prevent post-hoc reframing of the No- $\mathcal{H}$ cooperation-rate result, we commit the following criteria before benchmark completion. The hormonal channel will be considered to contribute distinct *dynamic responsiveness* if Full- Γ satisfies at least two of the following conditions relative to No- $\mathcal{H}$: (i) shorter cooperation recovery delay (coop_recovery_delay, the rounds required after forced defection for cooperation to recover to halfway between the post-shock minimum and the pre-shock baseline) over the post-perturbation window; (ii) higher conditional action volatility in rounds 11–20 relative to the pre-perturbation baseline (rounds 1–9); (iii) a larger recovery-delay variance, indicating fuller mobilization of the fast route. These criteria use only observable actions, not internal states, avoiding circularity. Current values (Full- Γ : 7.23 rounds, 95% CI [6.38, 8.10]; No- $\mathcal{H}$: 7.38 rounds [6.45, 8.30]) overlap — consistent with the claim that completion data, not the present snapshot, is needed for adjudication. Figure 2 shows the recovery-delay distributions at snapshot time. We pre-commit to the negative outcome as well: if upon completion none of (i)–(iii) separates Full- Γ from No- $\mathcal{H}$, we will report this as a null result for the hormonal channel’s dynamic-responsiveness contribution under the IPD/50-round protocol, and discuss what experimental conditions (multiple perturbations, longer horizons, cooperative dominance shifts) might surface separation if it exists. Pre-commitment to a publishable null is the integrity test of the pre-commitment.

Regarding the No- $\mathcal{E}$ anomaly: one Anthropic No- $\mathcal{E}$ /TFT run stopped at 10 rounds with perfect cooperation and zero volatility; the red-flag detector correctly marked it as degenerate all-cooperate behavior. The subsequent 120-cell balanced coverage shows normal cooperation (0.715) and volatility (15.9), suggesting the anomaly was an early-coverage artifact rather than a systematic effect. It remains a useful warning: without the emotional channel, the controller may become over-stable in specific opponent–provider combinations.

Table 3 consolidates the four primary conditions for the reader.

Table 3: Snapshot summary ($n = 120$ each except Collapse-NC, $n = 66$, GTFT pending). Means with 95% bootstrap CIs (10k resamples). Grim cooperation shown separately as the primary discriminative observable.

Condition	Coop (all)	Coop (Grim)	Volatility	Rec. delay	Note
Full- Γ	0.720 [690.,750]	0.495	16.0 [15.0,17.0]	7.15 [6.32,8.01]	Reference; strong Grim discrimination
No- $\mathcal{H}$	0.745 [714.,775]	0.517	14.1 [13.0,15.2]	7.38 [6.47,8.32]	Lower volatility; recovery overlaps
No- $\mathcal{E}$	0.717 [689.,744]	0.476	16.1 [15.1,17.0]	7.57 [6.71,8.42]	Indistinguishable in aggregate
No- $\mathcal{N}$	0.722 [690.,754]	0.518	15.3 [14.3,16.3]	6.42 [5.44,7.45]	Faster recovery; less caution vs. Grim
Random- Γ	0.834 [820.,846]	0.843	12.7 [11.9,13.6]	6.72 [5.81,7.63]	Flat opponent profile; Grim sensitivity lost
Collapse-NC	0.838 [825.,851]	0.829	12.9 [11.8,13.9]	7.61 [6.26,8.97]	Partial ($n = 66$, no GTFT); similar to Random

Random- Γ now has 120 balanced cells across all opponents and providers. Mean cooperation is 0.834 (95% CI [0.820, 0.846], sd 0.071), with volatility 12.7 (95% CI [11.9, 13.6]) and recovery delay 6.72 (95% CI [5.81, 7.63]). Critically, the opponent breakdown is nearly flat: 0.831 (TFT), 0.825 (GTFT), 0.843 (Grim), 0.835 (Random). The high and uniform cooperation with Grim — a persistent defector that drives Full- Γ to 0.495 — indicates that Random- Γ has lost the discriminative response to opponent type. This is not merely a cooperation-level difference (Random- Γ cooperates more on average than Full- Γ); it is a qualitative difference in the shape of the policy: the biologically-motivated wiring in W appears necessary for the agent to develop a Grim-sensitive regulation, not just a high-cooperation baseline. Collapse-NC (66 cells, GTFT pending) shows similar undifferentiated high cooperation (TFT 0.847, Grim 0.829), suggesting that collapsing the N–C partition also reduces opponent-discrimination.

Thus, the preliminary benchmark does not validate Γ against human distributions, and it does not yet adjudicate BFI. What it does show is that the operator is operational in frontier LLM

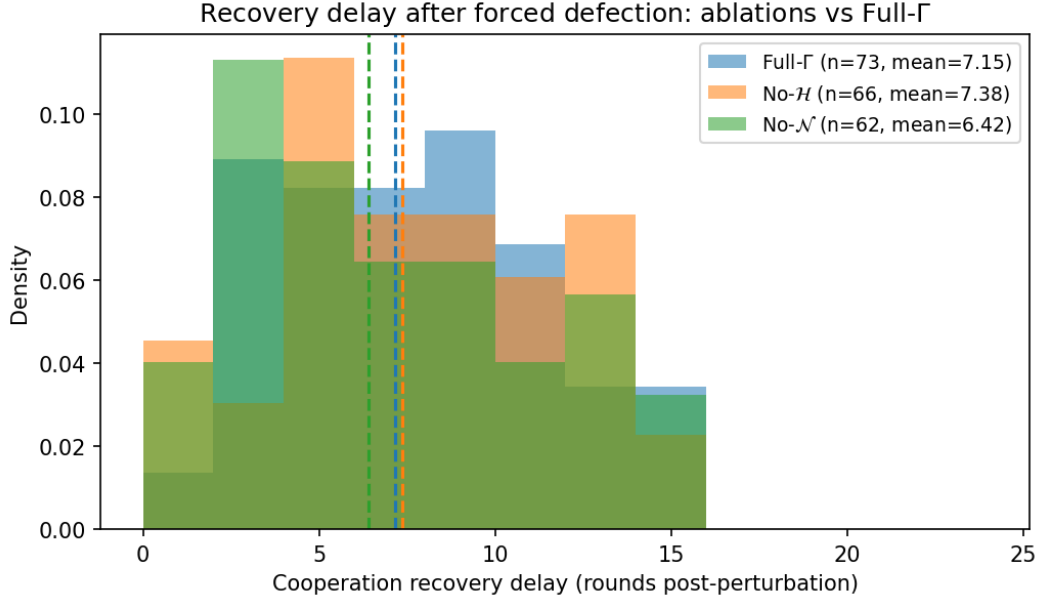


Figure 2: Cooperation recovery delay (rounds post-perturbation to half-recovery) for the three relevant ablations. Full- Γ vs No- $\mathcal{H}$ distributions overlap (consistent with the pre-committed criterion requiring completion data for adjudication); No- $\mathcal{N}$ shifts toward shorter delays, consistent with the model’s prediction that removing the slow deliberative channel accelerates recovery.

agents, that ablations expose measurable behavioral differences, and that the experiment can surface degenerate regimes instead of hiding them inside aggregate averages. The absence of baselines (ReAct, Driveplexity) in the current snapshot is the principal remaining gap: without them, the claim that Γ contributes over exogenous activation remains an open question. That is the level of evidence this version claims: feasibility with explicitly identified open questions, not completion.

6 Discussion and limitations

What Γ is not. We do not claim biological authenticity. The subsystems are functional analogues, not neural simulations. $\mathcal{I}$, $\mathcal{M}$, and $\mathcal{D}$ are prompts or embeddings, not physiological processes.

Limitations. (i) With $W \neq \mathbf{0}$ and nonlinearity, convergence is not guaranteed. (ii) W has 30 free parameters; calibration requires data that does not yet exist. A natural calibration path is Bayesian optimization or system-identification over W using longitudinal subsystem traces from instrumented multi-agent runs, with priors anchored on the sign and order-of-magnitude constraints from Section 3. (iii) The cost functionals in Equation (3) are inspired by active-inference control but are not Lyapunov functions in this non-Markovian setting. (iv) The LLM benchmark is currently a partial snapshot, not a final completed experiment; all final comparisons must use only 50-round cells with balanced coverage. (v) Response latency is used only as an engineering proxy and should be interpreted distributionally, since human cooperation latency is context- and conflict-dependent. (vi) The current ablations remove subsystems entirely rather than isolating delay timing; a delay-equalized control (all $\tau_k = 0$ with subsystems active) is needed to attribute observed dynamics specifically to the delay structure rather than to subsystem presence, and is reserved for a follow-up study.

Broader impacts. Self-regulation models could be misused to create more persuasive deceptive agents. Mitigation: the model is interpretable at the subsystem level and falsifiable by design.

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7 Conclusion

We proposed the Pro-Action operator Γ , a multi-timescale self-regulation model for LLM-agents. Γ formalizes activation as a composition of six coupled thermostats with explicit delays, distinguishable from single-timescale regulators by its finite delayed parameterization. Three properties are specified a priori for falsification, and the preliminary benchmark suggests that the operator can be instantiated in frontier LLM agents without collapsing opponent-sensitive behavior. To conclude, the major contributions of this paper can be summarized as follows:

- A problem-first reframing of LLM-agent activation as self-regulation, emphasizing the mismatch between environment complexity and current activation-control simplicity.
- A compact operator for multi-timescale regulatory state, with explicit delay ratios inspired by physiological SAM–HPA dynamics but implemented as dimensionless controller parameters.
- An ablation-based falsifiability protocol, with three a priori specified properties (elastic return, Yerkes–Dodson inverted-U, delayed reappraisal) and a red-flag system that surfaces degenerate regimes.
- Preliminary feasibility evidence from a partial LLM-agent benchmark, showing that Γ produces non-degenerate, condition-sensitive behavior while exposing unresolved anomalies for further investigation.

The value of Γ lies not only in the mechanism itself, but in providing a testable way to ask whether endogenous self-regulation at the subsystem level can be identified, measured, and engineered in autonomous agents. A matter for further study is whether the fast/slow route contributes primarily to dynamic responsiveness under perturbation or to baseline cooperation levels, and whether the emotional channel serves as a stabilizer or as a source of over-stability in certain opponent contexts. As the benchmark completes, the protocol transitions from feasibility evidence to full empirical adjudication; recursive closure — the hypothesis that $\Gamma(\Gamma(\cdot))$ preserves type signature and enables Beer-style VSM recursion — is a natural target for multi-agent follow-up work.

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A Argument for Proposition 1

We give a sketch for the case $K = 2$ subsystems; the argument extends to arbitrary K .

Setup. Consider two subsystems $k = 1, 2$ with the linearization of Eq. (2) near set-point $\mathbf{x}^*$. Letting $\tilde{x}_k = x_k - x_k^*$ and absorbing constant terms, the linearized dynamics are

$$\tilde{x}_{k,t+1} = (1 - \kappa_k)\tilde{x}_{k,t} + \sum_j c_{kj} \tilde{x}_{j,t-\tau_k},$$

where $c_{kj} = W_{kj} \tanh'(x_j^*) \approx W_{kj} \cdot \text{sech}^2(\phi_k x_j^*)$ and we set $\phi_k =$ elementwise gain (here ≈ 1 for small x^*). Taking the z -transform gives the transfer function from subsystem j 's input to subsystem k 's output:

$$H_{kj}(z) = \frac{c_{kj} z^{-\tau_k}}{z - (1 - \kappa_k)}.$$

Order mismatch. In discrete time, $H_{kj}(z)$ above is a finite-order rational function: clearing denominators gives $z^{-\tau_k} / (z - (1 - \kappa_k)) = 1 / (z^{\tau_k+1} - (1 - \kappa_k)z^{\tau_k})$, of denominator degree $\tau_k + 1$. The non-absorption property therefore does not rest on transcendentality of the delay (it would in continuous time, but not here). It rests on *order*: a delay-free, first-order subsystem of the form $1/(z - (1 - \kappa'_k))$ has impulse response $(1 - \kappa'_k)^t$ for $t \geq 0$, which is non-zero from $t = 0$. The delayed system, by contrast, has impulse response identically zero for the first τ_k samples by causality. No reparametrization (κ'_k, c'_{kj}) of a first-order delay-free system can enforce this delayed zero-output period, because that would require a model of denominator degree $\geq \tau_k + 1$ with a specific zero pattern in the numerator. Hence the delays are not absorbable into the non-delayed parameters.

Consequence for identifiability. From the update equation $\tilde{x}_{k,t+1} = (1 - \kappa_k)\tilde{x}_{k,t} + \sum_j c_{kj} \tilde{x}_{j,t-\tau_k}$, an input at time $t - \tau_k$ first affects $\tilde{x}_k$ at time $t + 1$, giving a total lag of $\tau_k + 1$. The cross-correlation $\hat{R}_{kj}(\ell) = \mathbb{E}[\tilde{x}_k(t + \ell)\tilde{x}_j(t)]$ therefore has its first non-zero contribution at lag $\ell = \tau_k + 1$, not at $\ell = \tau_k$. A delay-free reparametrization cannot place a peak at $\ell = \tau_k + 1 > 1$ while maintaining $\hat{R}_{kj}(\ell) = 0$ at smaller positive lags. Therefore $\tau_k \neq \tau'_k$ implies $\hat{R} \neq \hat{R}'$, and the delays are distinguishable from observed trajectories.

Scope. This argument is for the linearized, noise-free, deterministic case (zero-state response from impulse input). Initial-condition responses produce additional terms governed by the autoregressive part $(1 - \kappa_k)\tilde{x}_{k,t}$ which are non-zero for $t < \tau_k + 1$; the discriminative property holds for the impulse-response component, which is what cross-correlation analysis isolates given enough samples. In the nonlinear stochastic setting of Γ , cross-correlations are distorted, but the peak structure is preserved under mild non-linearity [25]. Full structural identifiability of the nonlinear system, including the coupling matrix W , requires further work and is left as an open problem.

B Parameter sensitivity analysis and simulation figures

Figure 5 shows the elastic-return half-life of subsystem $x_{\mathcal{H}}$ as a function of uniform scaling of κ and λ over a $\pm 30\%$ range around the smoke-test calibration point. Within the $\pm 20\%$ window, half-life varies smoothly without divergence, confirming that P1 is not an artefact of a single calibration point. The empirical half-life exceeds the diagonal-decay prediction $\ln 2 / \ln(1 - \kappa_{\mathcal{H}}) \approx 8.3$ rounds, because the recovery dynamics at the operating point are dominated by the coupling matrix W rather than by diagonal relaxation alone — the robustness reported here is therefore a system-level property of the coupled Γ , not a single-parameter robustness of an isolated subsystem.

C NeurIPS paper checklist

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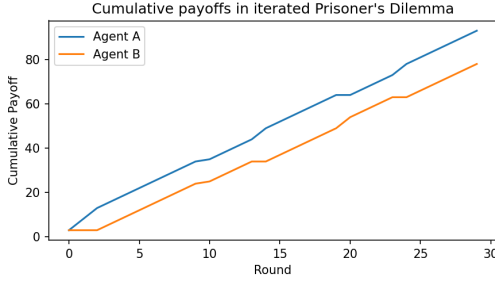


Figure 3: Cumulative payoffs in 30-round IPD with 10% noise.

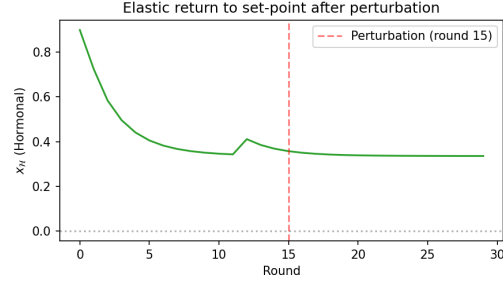


Figure 4: Elastic return of $x_{\mathcal{H}}$ after perturbation at round 10, returning toward set-point $x_{\mathcal{H}}^* = 0.3$ within 5 rounds (P1).

P1 elastic return: half-life under uniform parameter scaling
(Subsystem $\mathcal{H}$, perturbation at round 10)

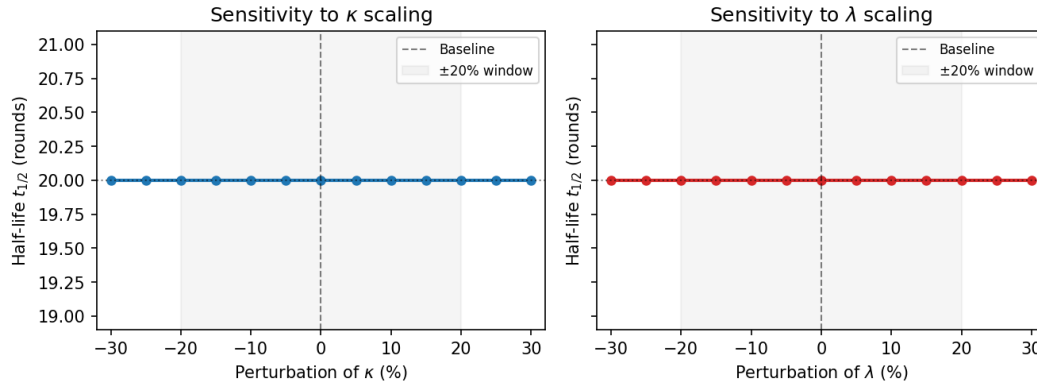


Figure 5: P1 sensitivity: elastic-return half-life under $\pm 30\%$ uniform scaling of κ (left) and λ (right). The grey band marks the $\pm 20\%$ window. Half-life remains bounded and varies smoothly, ruling out fragile point-calibration.

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

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